

Package ‘RobMixReg’

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Type Package

Title Robust Mixture Regression

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Description

Finite mixture models are a popular technique for modelling unobserved heterogeneity or to approximate general distribution functions in a semi-parametric way. They are used in a lot of different areas such as astronomy, biology, economics, marketing or medicine.

This package is the implementation of popular robust mixture regression methods based on different algorithms including: fleximix, finite mixture models and latent class regression; CTLERob, component-wise adaptive trimming likelihood estimation; mixbi, bi-square estimation; mixL, Laplacian distribution; mixt, t-distribution; TLE, trimmed likelihood estimation.

The implemented algorithms includes: CTLERob stands for Component-wise adaptive Trimming Likelihood Estimation based mixture regression; mixbi stands for mixture regression based on bi-square estimation; mixL stands for mixture regression based on Laplacian distribution; TLE stands for Trimmed Likelihood Estimation based mixture regression. For more detail of the algorithms, please refer to below references.

Reference: Chun Yu, Weixin Yao, Kun Chen (2017) <[doi:10.1002/cjs.11310](https://doi.org/10.1002/cjs.11310)>.

Neykov N, Filzmoser P, Dimova R et al. (2007) <[doi:10.1016/j.csda.2006.12.024](https://doi.org/10.1016/j.csda.2006.12.024)>.

Bai X, Yao W, Boyer JE (2012) <[doi:10.1016/j.csda.2012.01.016](https://doi.org/10.1016/j.csda.2012.01.016)>.

Wennan Chang, Xinyu Zhou, Yong Zang, Chi Zhang, Sha Cao (2020) <[doi:10.48550/arXiv.2005.11599](https://doi.org/10.48550/arXiv.2005.11599)>.

Depends R (>= 3.5.0)

License GPL

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LazyData true

Imports flexmix, robustbase, gtools, MASS,
methods, robust, lars, dplyr, rlang,
scales, ggplots, grDevices, graphics, RColorBrewer, stats

RoxygenNote 7.3.2

URL <https://changwn.github.io/RobMixReg/>

BugReports <https://github.com/changwn/RobMixReg/issues>

NeedsCompilation no

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biscalew	<i>biscalew : Robust M-estimates for scale.</i>
----------	---

Description

Tukey’s bisquare family of functions.

Usage

biscalew(t)

Arguments

t Numerical input, usually residuals.

Value

bisquare weight for scale.

bisquare	<i>bisquare : Robust estimates for mean.</i>
----------	--

Description

Tukey’s bisquare family of functions.

Usage

bisquare(t, k = 4.685)

Arguments

t Numerical input, usually residuals.
k A constant tuning parameter, default is 4.685.

Value

A bi-square weight for mean.

blockMap	<i>Plot the coefficient matrix.</i>
----------	-------------------------------------

Description

Plot the coefficient matrix.

Usage

```
blockMap(rrr)
```

Arguments

rrr	The result from CSMR function
-----	-------------------------------

calcobj_i	<i>calcobj_i: it calculates the objective function</i>
-----------	--

Description

function to calculate the objective function.

Usage

```
calcobj_i(X, Y, iter, pa)
```

Arguments

X	independent variable.
Y	response variable.
iter	see original CWM method.
pa	see original CWM method.

Value

see original CWM method.

CCLE_data

RobMixReg package built-in CCLE data.

Description

The list which contain all the information to generate variables used in the real application.

Usage

```
CCLE_data
```

Format

A list whose length is 2:

X Gene expression dataset.

Y AUCC score.

colon_data

RobMixReg package built-in Colon cancer data.

Description

The list which contain all the information to generate variables used in the real application.

Usage

```
colon_data
```

Format

A list whose length is 3:

rnames A string contains the name of binding protein and epigenetic regulator.

x3 The gene expression profile of CREB3L1.

y3 The methylation profile of cg16012690 on 299 colon adenocarcinoma patients.

x2 x2

y2 y2

x1 x1

y1 y1

compPlot	<i>The plot wrapper function.</i>
----------	-----------------------------------

Description

The plot wrapper function.

Usage

```
compPlot(type = "rlr", x, y, nc, inds_in, res)
```

Arguments

type	The character to choose which type of plot to generate.
x	The independent variables
y	The external variable
nc	The number of components
inds_in	A vector indicate the outlier samples.
res	The result object returned by MLM function.

Compute_Rbase_SVD	<i>Compute the row space using SVD.</i>
-------------------	---

Description

Compute the row space using SVD.

Usage

```
Compute_Rbase_SVD(bulk_data, tg_R1_lists_selected)
```

Arguments

bulk_data	The bulk data..
tg_R1_lists_selected	A list of the marker genes for several cell types.

Value

A matrix which each row span the row space using cell type specific marker genes.

CSMR*The main function of the RBSL algorithm.*

Description

The main function of the RBSL algorithm.

Usage

```
CSMR(x, y, nit, nc, max_iter)
```

Arguments

x	The matrix
y	The external supervised variable.
nit	xxx?
nc	The component number in the mixture model.
max_iter	The maximum iteration number.

Value

A list object consist of coefficient, clustering membership, data x, external variable y, predicted y based on regression model.

CSMR_one*Perform the RBSL algorithm one times.*

Description

Perform the RBSL algorithm one times.

Usage

```
CSMR_one(x, y, nit = 1, nc, max_iter)
```

Arguments

x	The matrix
y	The external supervised variable.
nit	xxx?
nc	The component number in the mixture model.
max_iter	The maximum iteration number.

Value

A list object consist of coefficient, clustering membership, data x, external variable y, predicted y based on regression model.

CSMR_predict	<i>The predict function of the CSMR algorithm.</i>
--------------	--

Description

The predict function of the CSMR algorithm.

Usage

```
CSMR_predict(CSMR_coffs, CSMR.model, xnew, ynew, singleMode = F)
```

Arguments

CSMR_coffs	The coefficient matrix.
CSMR.model	The trained model.
xnew	x variable.
ynew	y variable.
singleMode	A parameter to set the component to one.

Value

A list object consist of coefficient, clustering membership, data x, external variable y, predicted y based on regression model.

CSMR_train	<i>The train function of the CSMR algorithm.</i>
------------	--

Description

The train function of the CSMR algorithm.

Usage

```
CSMR_train(x, y, nit, nc, max_iter)
```

Arguments

x	The matrix
y	The external supervised variable.
nit	xxx
nc	The component number in the mixture model.
max_iter	The maximum iteration number.

Value

A list object consist of coefficient, clustering membership, data x, external variable y, predicted y based on regression model.

CTLRob	<i>CTLRob: Robust mixture regression based on component-wise adaptive trimming likelihood estimation.</i>
--------	---

Description

CTLRob performs robust linear regression with high breakdown point and high efficiency in each mixing components and adaptively remove the outlier samples.

Usage

```
CTLRob(formula, data, nit = 20, nc = 2, rlr_method = "ltsReg")

## S4 method for signature 'formula,ANY,ANY,numeric'
CTLRob(formula, data, nit = 20, nc = 2, rlr_method = "ltsReg")
```

Arguments

formula	A symbolic description of the model to be fit.
data	A data frame containing the predictor and response variables, where the last column is the response variable.
nit	Number of iterations.
nc	Number of mixture components.
rlr_method	The regression methods, default is 'ltsReg'.

denLp	<i>denLp : Density function for Laplace distribution.</i>
-------	---

Description

Laplace distribution.

Usage

```
denLp(rr, sig)
```

Arguments

rr	Shift from the location parameter
sig	Scale parameter.

Value

Laplace density.

DeOut	<i>DeOut : Detect outlier observations.</i>
-------	---

Description

Detect outlier observations from a vector.

Usage

```
DeOut(daData, method)
```

Arguments

daData	A numerical vector.
method	Choose from '3sigma', 'hampel' and 'boxplot'.

Value

indices of outlier observations.

dmnorm	<i>dmnorm: multivariate normal density function</i>
--------	---

Description

see original CWM method.

Usage

```
dmnorm(X, mu, sigma)
```

Arguments

X	see original CWM method.
mu	see original CWM method.
sigma	see original CWM method.

Value

see original CWM method.

estimClustPar_i	<i>estimClustPar_i: see original CWM method</i>
-----------------	---

Description

see original CWM method.

Usage

estimClustPar_i(X, Y, iter, pa)

Arguments

X	independent variables.
Y	response variable.
iter	see original CWM method.
pa	see original CWM method.

Value

xsee original CWM method.

findClustAssig_i	<i>findClustAssig_i: it obtains the assignment corresponding to each point to the K populations and the set of trimmed points</i>
------------------	---

Description

see original CWM method.

Usage

findClustAssig_i(X, Y, iter, pa)

Arguments

X	independent variable.
Y	response variable.
iter	see original CWM method.
pa	see original CWM method.

Value

see original CWM method.

flexmix_2	<i>flexmix_2: Multiple runs of MLE based mixture regression to stabilize the output.</i>
-----------	--

Description

Mixture regression based on MLE could be unstable when assuming unequal variance. Multiple runs of flexmix is performed to stabilize the results.

Usage

```
flexmix_2(formula, data1, k, mprior)
```

Arguments

formula	A symbolic description of the model to be fit.
data1	A data frame containing the predictor and response variables, where the last column is the response variable.
k	Number of mixture components.
mprior	A numeric number in (0,1) that specifies the minimum proportion of samples in each mixing components.

Value

A S4 object of flexmix class.

gaussData	<i>RobMixReg package built-in gaussian example data.</i>
-----------	--

Description

A dataset generated from gaussian distribution in RobMixReg package.

Usage

```
gaussData
```

Format

A data frame with 100 rows and 3 variables:

x	x variable
y	y variable
c	cluster information

getini	<i>getini: see original CWM method</i>
--------	--

Description

see original CWM method.

Usage

```
getini(K, n)
```

Arguments

K	see original CWM method.
n	see original CWM method.

Value

see original CWM method.

InitClusters_i	<i>InitClusters_i: see original CWM method</i>
----------------	--

Description

see original CWM method.

Usage

```
InitClusters_i(X, Y, iter, pa)
```

Arguments

X	independent variable.
Y	response variable.
iter	see original CWM method.
pa	see original CWM method.

Value

see original CWM method.

lars.lsa	<i>lars variant for LSA.</i>
----------	------------------------------

Description

lars variant for LSA.

Usage

```
lars.lsa(  
  Sigma0,  
  b0,  
  intercept,  
  n,  
  type = c("lasso", "lar"),  
  eps = .Machine$double.eps,  
  max.steps  
)
```

Arguments

Sigma0	The parameter.
b0	The intercept of the regression line.
intercept	The bool variable of whether consider the intercept situation
n	The number of data point.
type	Regression options, choose form "lasso" or "lar".
eps	The converge threshold defined by the machine.
max.steps	The maximum iteration times to stop.

Value

object.

Author(s)

Reference Wang, H. and Leng, C. (2006) and Efron et al. (2004).

logLik_mixtureReg	<i>Obtain Log-likelihood from a mixtureReg Object</i>
-------------------	---

Description

S3 method for class 'mixtureReg'. However, it doesn't return a 'logLik' object. For simplicity, it returns a 'numeric' value.

Usage

```
logLik_mixtureReg(mixtureModel)
```

Arguments

mixtureModel mixtureReg object, typically result from 'mixtureReg()'.

Value

Return a numeric value of log likelihood.

lsa	<i>Least square approximation. This version Oct 19, 2006.</i>
-----	---

Description

Least square approximation. This version Oct 19, 2006.

Usage

```
lsa(obj)
```

Arguments

obj lm/glm/coxph or other object.

Value

beta.ols: the MLE estimate ; beta.bic: the LSA-BIC estimate ; beta.aic: the LSA-AIC estimate.

Author(s)

Reference Wang, H. and Leng, C. (2006) and Efron et al. (2004).

mixlinrb_bi	<i>mixlinrb_bi: mixlinrb_bione estimates the mixture regression parameters robustly using bisquare function based on multiply initial value.</i>
-------------	--

Description

An EM-type of parameter estimation by replacing the least square estimation in the M-step with a robust criterion.

Usage

```
mixlinrb_bi(formula, data, nc = 2, nit = 200)

## S4 method for signature 'formula,ANY,numeric,numeric'
mixlinrb_bi(formula, data, nc = 2, nit = 20)
```

Arguments

formula	A symbolic description of the model to be fit.
data	A data frame containing the predictor and response variables, where the last column is the response variable.
nc	Number of mixture components.
nit	Number of iterations for biSauqre method.

Value

Estimated coefficients of all components.

mixlinrb_bione	<i>mixlinrb_bione : mixlinrb_bione estimates the mixture regression parameters robustly using bisquare function based on one initial value.</i>
----------------	---

Description

An EM-type of parameter estimation by replacing the least square estimation in the M-step with a robust criterion.

Usage

```
mixlinrb_bione(formula, data, nc = 2)
```


Arguments

formula	A symbolic description of the model to be fit.
data	A data frame containing the predictor and response variables, where the last column is the response variable.
nc	Number of mixture components.

Value

Estimated coefficients of all components.

mixLp	<i>mixLp : mixLp_one estimates the mixture regression parameters robustly using Laplace distribution based on multiply initial value..</i>
-------	--

Description

mixLp estimates the mixture regression parameters robustly using bisquare function based on multiple initial values. The solution is found by the modal solution.

Usage

```
mixLp(formula, data, nc=2, nit=200)

## S4 method for signature 'formula,ANY,numeric,numeric'
mixLp(formula, data, nc = 2, nit = 20)
```

Arguments

formula	A symbolic description of the model to be fit.
data	A data frame containing the predictor and response variables, where the last column is the response variable.
nc	Number of mixture components.
nit	Number of iterations

Value

Estimated coefficients of all components.

Examples

```
library("RobMixReg")
formula01=as.formula("y~x")
x=(gaussData$x);y=as.numeric(gaussData$y);
example_data01=data.frame(x,y)
res = mixLp(formula01, example_data01, nc=2, nit=20)
```

mixLp_one	<i>mixLp_one : mixLp_one estimates the mixture regression parameters robustly using Laplace distribution based on one initial value.</i>
-----------	--

Description

Robust mixture regression assuming that the error terms follow a Laplace distribution.

Usage

```
mixLp_one(formula, data, nc = 2)
```

Arguments

formula	A symbolic description of the model to be fit.
data	A data frame containing the predictor and response variables, where the last column is the response variable.
nc	Number of mixture components.

Value

Estimated coefficients of all components.

mixtureReg	<i>Function to Fit Mixture of Regressions</i>
------------	---

Description

The main function in this package.

Usage

```
mixtureReg(
  regData,
  formulaList,
  xName = NULL,
  yName = NULL,
  mixingProb = c("Constant", "loess"),
  initialWList = NULL,
  epsilon = 1e-08,
  max_iter = 10000,
  max_restart = 15,
  min_lambda = 0.01,
  min_sigmaRatio = 0.1,
  silently = TRUE
)
```

Arguments

<code>regData</code>	data frame used in fitting model.
<code>formulaList</code>	a list of the regression components that need to be estimated.
<code>xName</code>	character; Name used to pick x variable from data.
<code>yName</code>	character; Name used to pick y variable from data.
<code>mixingProb</code>	character; Specify how the mixing probabilities are estimated in the M step. "Constant" specifies a constant mixing probabilities; "loess" specifies predictor dependent mixing probabilities obtained by loess smoothing.
<code>initialWList</code>	a list of weights guesses (provided by user). Typically this is not used, unless the user has a good initial guess.
<code>epsilon</code>	a small value that the function consider as zero. The value is used in determine matrix singularity and in determine convergence.
<code>max_iter</code>	the maximum number of iterations.
<code>max_restart</code>	the maximum number of restart before giving up.
<code>min_lambda</code>	a value used to ensure estimated mixing probabilities (lambda's) are not too close to zero.
<code>min_sigmaRatio</code>	a value used to prevent estimated variaces of any regression component from collapsing to zero.
<code>silently</code>	a switch to turn off the screen printout.

Value

A class 'mixtureReg' object.

Author(s)

The mixtureReg package is developed by Tianxia Zhou on github. All right reserved by Tianxia Zhou.

MLM

The main function of mining the latent relationship among variables.

Description

The main function of mining the latent relationship among variables.

Usage

```
MLM(
  ml.method = "rlr",
  rmr.method = "cat",
  b.formulaList = list(formula(y ~ x), formula(y ~ 1)),
  formula = y ~ x,
```

```

nc = 2,
nit = 1,
x = NULL,
y = NULL,
max_iter = 50,
tRatio = 0.05
)

```

Arguments

ml.method	The option to select the four methods in vignette.
rmr.method	The option to select the robust mixture regression method.
b.formulaList	The case b require the user provide the formula list. This enable the flexible mixture regression.
formula	The linear relationship between two variables.
nc	Number of mixture components.
nit	Number of iterations for CTLE, mixbi, mixLp.
x	The matrix x of the high dimension situation.
y	The external outcome variable.
max_iter	Maximum iteration for TLE method.
tRatio	The ratio of the outliers in the TLE robust mixture regression method.

Value

Main result object.

MLM_bic	<i>Model selection function for low dimension data.</i>
---------	---

Description

Model selection function for low dimension data.

Usage

```
MLM_bic(ml.method = "rlr", x, y, nc = 1, formulaList = NULL, K = 2)
```

Arguments

ml.method	The parameter to choose the fitted model for calculating the BIC
x	x variable.
y	y variable.
nc	The component number for low dimensional feature
formulaList	The list of target formular
K	The component number for high dimensional feature

Value

BIC value.

MLM_cv

Cross validation (fold-5) function for high dimension data.

Description

Cross validation (fold-5) function for high dimension data.

Usage

```
MLM_cv(x = NULL, y = NULL, nit = 1, nc = 2, max_iter = 50)
```

Arguments

x	x variable.
y	y variable.
nit	Iteration number.
nc	The number of component.
max_iter	Maximum iteration.

Value

The correlation between y and y_hat based on five fold cross validation.

orderedLines

Sort by X Coordinates and Add Line to a Plot

Description

Rearrange X and Y coordinates before calling "lines()" function.

Usage

```
orderedLines(x, y, ...)
```

Arguments

x	X coordinate vectors of points to join.
y	Y coordinate vectors of points to join.
...	Further graphical parameters.

plot_CTLE	<i>plot_CTLE: Plot the mixture/single regression line(s) in a simply function.</i>
-----------	--

Description

CTLERob performs robust linear regression with high breakdown point and high efficiency in each mixing components and adaptively remove the outlier samples.

Usage

```
plot_CTLE(formula, data, nc = 2, inds_in)

## S4 method for signature 'formula,ANY,numeric'
plot_CTLE(formula, data, nc = 2, inds_in)
```

Arguments

formula	A symbolic description of the model to be fit.
data	A data frame containing the predictor and response variables, where the last column is the response variable.
nc	Number of mixture components.
inds_in	The index of the point which belongs to the current regression line.

plot_mixtureReg	<i>Plot Fit and Mixing Probability of a mixtureReg Object</i>
-----------------	---

Description

S3 plot method for class 'mixtureReg'.

Usage

```
plot_mixtureReg(
  mixtureModel,
  which = 1:2,
  xName = NULL,
  yName = NULL,
  xlab = NULL,
  ylab = NULL,
  ...
)
```

Arguments

mixtureModel	mixtureReg object, typically result from 'mixtureReg()'.
which	numeric; choose which plot to display. '1' gives a plot of fit; '2' gives a plot of mixing probability.
xName	character; Name used to pick x variable from data.
yName	character; Name used to pick y variable from data.
xlab	character; label that should be put on the x axis.
ylab	character; label that should be put on the y axis.
...	Further graphical parameters.

plot_mixtureRegList *Plot a List of mixtureReg Objects*

Description

Feed in a list of mixtureReg models and get an overlayed plot.

Usage

```
plot_mixtureRegList(mixtureRegList, xName = NULL, yName = NULL, ...)
```

Arguments

mixtureRegList	a list of multiple mixtureReg objects.
xName	character; Name used to pick x variable from data.
yName	character; Name used to pick y variable from data.
...	Further graphical parameters.

Rec_Lm *Adaptive lasso.*

Description

Adaptive lasso.

Usage

```
Rec_Lm(XX, yy)
```

Arguments

XX	The independent variable.
yy	The dependent variable.

Value

A list object consist of index of selected variable and coefficient for all variables.

restr.diffax_i1	<i>restr.diffax_i1: see original CWM method</i>
-----------------	---

Description

see original CWM method.

Usage

restr.diffax_i1(iter, pa)

Arguments

- iter see original CWM method.
- pa see original CWM method.

Value

see original CWM method.

restr.diffax_i2	<i>restr.diffax_i2: see original CWM method</i>
-----------------	---

Description

see original CWM method.

Usage

restr.diffax_i2(iter, pa)

Arguments

- iter see original CWM method.
- pa see original CWM method.

Value

see original CWM method.

restr2_eigenv	<i>restr2_eigenv: see original CWM method</i>
---------------	---

Description

see original CWM method.

Usage

```
restr2_eigenv(autovalues, ni.ini, factor, zero.tol)
```

Arguments

autovalues	see original CWM method.
ni.ini	see original CWM method.
factor	see original CWM method.
zero.tol	see original CWM method.

Value

see original CWM method.

rnr	<i>The main function of Robust Mixture Regression using five methods.</i>
-----	---

Description

The main function of Robust Mixture Regression using five methods.

Usage

```
rnr(
  lr.method = "flexmix",
  formula = NULL,
  data = NULL,
  nc = 2,
  nit = 20,
  tRatio = 0.05,
  MaxIt = 200
)
```

Arguments

<code>lr.method</code>	A robust mixture regression method to be used. Should be one of "flexmix", "TLE", "CTLERob", "mixbi", "mixLp".
<code>formula</code>	A symbolic description of the model to be fit.
<code>data</code>	A data frame containing the predictor and response variables, where the last column is the response variable.
<code>nc</code>	Number of mixture components.
<code>nit</code>	Number of iterations for CTLE, mixbi, mixLp.
<code>tRatio</code>	Trimming proportion for TLE method.
<code>MaxIt</code>	Maximum iteration for TLE method.

Value

An S4 object about the regression result.

Examples

```
library(RobMixReg)
#library(robust)
library(flexmix)
library(robustbase)
library(MASS)
library(gtools)
# gaussData
x=(gaussData$x);y=as.numeric(gaussData$y);
formula01=as.formula("y~x")
example_data01=data.frame(x,y)
res_rmr = rmr(lr.method='flexmix', formula=formula01, data=example_data01)
res_rmr = rmr(lr.method='CTLERob', formula=formula01, data=example_data01)
```

RobMixReg-class

Class RobMixReg.

Description

Class RobMixReg defines a robust mixture regression class as a S4 object.

Slots

`inds_in` The indices of observations used in the parameter estimation.
`indout` The indices of outlier samples, not used in the parameter estimation.
`ctleclusters` The cluster membership of each observation.
`compcoef` Regression coefficients for each component.
`comppvals` Component p values.
`compwww` The posterior of the clustering.
`call` Call function.

simuData	<i>RobMixReg package built-in simulated example data.</i>
----------	---

Description

A simulation dataset from RobMixReg package. This simulation dataset is in dimension 2 and ground truth (include outliers label) of the cluster information also generated.

Usage

```
simuData
```

Format

A data frame with 500 rows and 5 variables:

X1 X1 variable
X2 X2 variable
y y variable
c cluster information
outlier outlier indicator

simu_data_sparse	<i>Simulate high dimension data for RBSL algorithm validation.</i>
------------------	--

Description

Simulate high dimension data for RBSL algorithm validation.

Usage

```
simu_data_sparse(n, bet, pr, sigma)
```

Arguments

n	Patient number.
bet	The coefficient matrix.
pr	A vector of probability threshold which simulate the sampling based on uniform distribution.
sigma	A vector of noise level. The length should be equal to the component number.

Value

A list object consist of x, y, true cluster label.

simu_func	<i>The simulation function for low/high dimensional space.</i>
-----------	--

Description

The simulation function for low/high dimensional space.

Usage

```
simu_func(beta, sigma, alpha = NULL, n = 400)
```

Arguments

beta	The slope vector for low dimensional space or matrix for high dimensional space.
sigma	A vector whose k-th element is the standard deviation for the k-th regression component.
alpha	The parameter to control the number of outliers for low dimensional space.
n	The sample number for high dimensional data.

Value

A list object.

simu_low	<i>The simulation function for low dimensional space.</i>
----------	---

Description

The simulation function for low dimensional space.

Usage

```
simu_low(beta, inter, alpha = NULL)
```

Arguments

beta	The slope vector.
inter	The intercept vector.
alpha	The parameter to control the number of outliers.

Value

A list object consists of the x variable in low dimensional space and the external y variable.

ssclmat	<i>ssclmat: see original CWM method</i>
---------	---

Description

see original CWM method.

Usage

```
ssclmat(x, k)
```

Arguments

x	see original CWM method.
k	see original CWM method.

Value

see original CWM method.

TLE	<i>TLE: robust mixture regression based on trimmed likelihood estimation.</i>
-----	---

Description

The algorithm fits a mixture regression model after trimming a proportion of the observations, given by tRatio.

Usage

```
TLE(formula, data, nc = 2, tRatio, MaxIt = 200)
```

```
## S4 method for signature 'formula,ANY,numeric,numeric,numeric'
TLE(formula, data, nc = 2, tRatio, MaxIt = 200)
```

Arguments

formula	A symbolic description of the model to be fit.
data	A data frame containing the predictor and response variables, where the last column is the response variable.
nc	Number of mixture components.
tRatio	Trimming proportion.
MaxIt	Maximum iteration.

Value

A S4 object of RobMixReg class.

Examples

```
library("RobMixReg")
formula01=as.formula("y~x")
x=(gaussData$x);y=as.numeric(gaussData$y);
example_data01=data.frame(x,y)

res = TLE(formula01,example_data01, nc=2,tRatio=0.05,MaxIt=200)
```

TreatSingularity	<i>TreatSingularity: see original CWM method</i>
------------------	--

Description

see original CWM method.

Usage

```
TreatSingularity(iter, pa)
```

Arguments

- iter see original CWM method.
- pa see original CWM method.

Value

see original CWM method.

trim.cwm	<i>trim.cwm: main function</i>
----------	--------------------------------

Description

main function adapted from the original developer of CWM method.

Usage

```
trim.cwm(  
  X,  
  Y,  
  K,  
  niter = 20,  
  Ksteps = 10,  
  maxfactx = 5,  
  maxfacty = 5,  
  zero.tol = 1e-16,  
  zero.tol1 = 1e-16,  
  zero.tol2 = 1e-90,  
  trace = 0,  
  alpha = 0.1,  
  init = 5  
)
```

Arguments

X	independent variables.
Y	response variable.
K	number of clusters.
niter	see original CWM method.
Ksteps	see original CWM method
maxfactx	see original CWM method
maxfacty	see original CWM method
zero.tol	see original CWM method
zero.tol1	see original CWM method
zero.tol2	see original CWM method
trace	see original CWM method
alpha	see original CWM method
init	number of initializations

Value

an RobMixReg object.

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